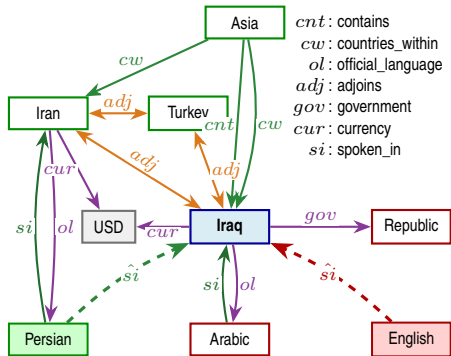


(a) Evidence Motifs

query = $\langle ?, \text{spoken_in}, \text{Iraq} \rangle$



(b) Ground Truth: Persian

Rank: 2 $\rightarrow$ 1
Score: 0.260 $\rightarrow$ 0.301

Rules (#Rule=109)

- ϕ_1 : $si(X,Y) \leftarrow si(X,A) \wedge adj(A,Y)$ [0.249]
- ϕ_2 : $si(X,Y) \leftarrow si(X,A) \wedge cw(B,A) \wedge cnt(B,Y)$ [0.060]
- ϕ_3 : $si(X,Y) \leftarrow ol(A,X) \wedge adj(A,B) \wedge adj(Y,B)$ [0.179]
- ϕ_4 : $si(X,Y) \leftarrow ol(A,X) \wedge cur(A,B) \wedge cur(Y,B)$ [0.068]
- ϕ_5 : $si(Persian,Y) \leftarrow cnt(Asia,Y)$ [0.093]

Dependency (#Complementarity=156)

- $d_{1,2}$: (ϕ_1, ϕ_2) [gain= +0.026]
- $d_{3,4}$: (ϕ_3, ϕ_4) [gain= +0.014]

$$\Delta_{dep} = +0.041$$

adjacency paths and regional paths reinforce Persian

(c) False Positive: English

Rank: 1 $\rightarrow$ 3
Score: 0.300 $\rightarrow$ 0.178

Rules (#Rule=171)

- ϕ_6 : $si(English,Y) \leftarrow gov(Y,Republic)$ [0.194]
- ϕ_7 : $si(English,Y) \leftarrow gov(Y,A)$ [0.253]
- ϕ_8 : $si(English,Y) \leftarrow cnt(Asia,Y)$ [0.222]
- ϕ_9 : $si(English,Y) \leftarrow cw(Asia,Y)$ [0.267]
- ϕ_{10} : $si(English,Y) \leftarrow si(Arabic,Y)$ [0.318]

Dependency (#Redundancy=201)

- $d_{6,7}$: (ϕ_6, ϕ_7) [gain= -0.038]
- $d_{8,9}$: (ϕ_8, ϕ_9) [gain= -0.018]

$$\Delta_{dep} = -0.122$$

government and continent cues overlap for English